Postdoctoral Research Associate - Coastal Landscape Geospatial Modeling

Oak Ridge National Laboratory · Oak Ridge, TN

Job ID (this archive)	ENERGY-122
Employer	Oak Ridge National Laboratory
Location(s)	Oak Ridge, TN
Career stage (archive tag)	Graduate degree / research (Postdoctoral)
Location group	Other U.S.
Majors (archive tags)	Computational & Geospatial Science
Energy area	DOE earth-system science / remote sensing
Pay listed	Not listed
Recruiting status when checked	Status not confirmed: the posting text was found, but current hiring was not verified. Checked Oct 26, 2026.
Source page	https://jobs.ornl.gov/job/Oak-Ridge-Postdoctoral-Research-Associate-Coastal-Landscape-Geospatial-Modeling-TN-37830/1431014900/
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-093

Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

Requisition Id 17130

Overview:

The Plant-Soil Interactions Group within the Environmental Sciences Division (ESD) at Oak Ridge National Laboratory (ORNL) seeks a postdoctoral researcher in Coastal Landscape analysis and modeling, with a focus on training and application of predictive machine learning tools. This researcher will synthesize regional-scale remote sensing imagery and ecological monitoring data and develop data-driven methods for predicting land loss and ecosystem transitions in wetland-rich landscapes of the Gulf Coast with a focus on coastal Louisiana.

This candidate will directly support the Exploring Gulf Region Ecosystem Transitions (EGRET) project, an interdisciplinary and multi-institutional collaboration focused on disturbance-driven ecosystem transitions and their impacts across the U.S. Gulf Coast. EGRET employs an integrated model–experiment (ModEx) approach accelerated by artificial intelligence (AI) to advance predictive understanding of how plant–microbial–soil interactions vary across inundation and salinity gradients to shape ecological, hydrological, and geomorphological responses to abrupt disturbance. A cohort of postdoctoral researchers will be hired across multiple institutions to collaboratively support scientific advances guided by AI/ML, remote sensing, process-based modeling, field observations and experiments, and advanced analytical techniques.

Major Duties/Responsibilities:

- Synthesize, harmonize, and analyze large environmental monitoring and remote sensing datasets from coastal Louisiana
- Train and apply multimodal and generative AI methods to predict land loss and vegetation change at high spatial resolution
- Work closely with remote-sensing scientists, modelers, and empiricists across DOE laboratories and universities to address project objectives.
- Present results within the research group and at national and international conferences and publish manuscripts in peer-reviewed journals.
- Deliver ORNL's mission by aligning behaviors, priorities, and interactions with our core values of Impact, Integrity, Teamwork, Safety, and Service.

Basic Qualifications:

- A PhD in Earth science, Ecology, Geography, Environmental Sciences, Geosciences, or a related field completed within the last 5 years or expected to be completed soon.
- Expertise in AI/ML methods in an environmental or Earth science context
- Experience programming in Python or related languages
- A strong record of independent and productive research, as indicated by conference presentations and/or publications in peer-reviewed journals.
- Ability to work independently and collaboratively as part of a large multidisciplinary team.
- Excellent interpersonal, oral, and written communication skills.

Preferred Qualifications:

- Experience with AI/ML approaches for spatiotemporal or Earth system data, including generative AI methods.
- Expertise in coastal landscape ecology, hydrology, and/or geomorphology.
- Experience analyzing large, heterogeneous environmental, geospatial, remote-sensing, or time-series datasets.

Contact for Position: Ben Sulman (sulmanbn@ornl.gov)

Special Requirements:

Applicants cannot have received their Ph.D. more than five years prior to the date of application and must complete all degree requirements before starting their appointment. The appointment length will be up to 24 months with the potential for extension. Initial appointments and extensions are subject to performance and availability of funding.

- Please submit three letters of reference when applying to this position. You may upload these directly to your application or have them sent to Postdocrecruitment@ornl.gov with the position title and number referenced in the subject line.
- Instructions to upload documents to your candidate profile:
- Login to your account via jobs.ornl.gov
- View Profile
- Under the My Documents section, select Add a Document
- For employment at Oak Ridge National Laboratory (ORNL), a Real ID compliant form of identification will be required. Additionally, ORNL is subject to Department of Energy (DOE) access restrictions. All employees must also be able to obtain and maintain a federal Personal Identity Verification (PIV) card as mandated by Homeland Security Presidential Directive 12 (HSPD-12) and Department of Energy (DOE) Order 473.1A, which requires a favorable post-employment background investigation.

To obtain this credential, new employees must successfully complete and pass a Federal Tier 1 background check investigation. This investigation includes a declaration of illegal drug activities, including use, supply, possession, or manufacture within the last year. This includes marijuana and cannabis derivatives, which are still considered illegal under federal law, regardless of state laws.

For foreign national candidates:

If you have not resided in the U.S. for three consecutive years, you are not eligible for the PIV credential and instead will need to obtain a favorable Local Site Specific Only (LSSO) risk determination to maintain employment. Once you meet the three-year residency requirement, you will be required to obtain a PIV credential to maintain employment.

About ORNL:

As a U.S. Department of Energy (DOE) Office of Science national laboratory, ORNL has an impressive 80-year legacy of addressing the nation’s most pressing challenges. Our team is made up of over 7,000 dedicated and innovative individuals! Our goal is to create an environment where a variety of perspectives and backgrounds are valued, ensuring ORNL is known as a top choice for employment. These principles are essential for supporting our broader mission to drive scientific breakthroughs and translate them into solutions for energy, environmental, and security challenges facing the nation.

ORNL offers competitive pay and benefits programs to attract and retain individuals who demonstrate exceptional work behaviors. The laboratory provides a range of employee benefits, including medical and retirement plans and flexible work hours, to support the well-being of you and your family. Employee amenities such as on-site fitness, banking, and cafeteria facilities are also available for added convenience.

Other benefits include the following: Prescription Drug Plan, Dental Plan, Vision Plan, 401(k) Retirement Plan, Contributory Pension Plan, Life Insurance, Disability Benefits, Generous Vacation and Holidays, Parental Leave, Legal Insurance with Identity Theft Protection, Employee Assistance Plan, Flexible Spending Accounts, Health Savings Accounts, Wellness

Programs, Educational Assistance, Relocation Assistance, and Employee Discounts.

#LI-DC1

This position will remain open for a minimum of 5 days after which it will close when a qualified candidate is identified and/or hired.

We accept Word (.doc, .docx), Adobe (unsecured .pdf), Rich Text Format (.rtf), and HTML (.htm, .html) up to 5MB in size. Resumes from third party vendors will not be accepted; these resumes will be deleted and the candidates submitted will not be considered for employment.

If you have trouble applying for a position, please email ORNLRecruiting@ornl.gov.

ORNL is an equal opportunity employer. All qualified applicants, including individuals with disabilities and protected veterans, are encouraged to apply. UT-Battelle is an E-Verify employer.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official posting description retrieved; application acceptance not independently established. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: A PhD in Earth science, Ecology, Geography, Environmental Sciences, Geosciences, or a related field completed within the last 5 years or expected to be completed soon.

Key skills listed in the archive: Python; machine learning; time-series; remote sensing

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Unverified · PDF: Complete · Source: Official employer careers Employer / ATS identifier: 17130

Career stage: Postdoctoral

Major fit: Advanced research pathway: coastal remote sensing and geospatial machine learning; PhD required. Energy-sector connection: DOE earth-system science; environmental research adjacent to the energy sector, not an energy-generation role.

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Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer careers. Source identifier: 17130.

Sources and retrieval details

Employer posting

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Retrieved: 2026-10-26T16:37:10.148190+00:00 (UTC). Major fit, career stage and energy-sector connection are archive annotations; the employer text above is preserved from the cited source.